

MRIDUL KUMAR

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EDUCATION

Indian Institute of Technology, Patna

B.Tech in Artificial Intelligence and Data Science

NOV 2022 – MAY 2026

CPI: 8.54

EXPERIENCE

RapidFort Inc.

JAN 2026 – PRESENT

Software Engineer (Docker, Kubernetes, Helm, Python, Bash, Linux, SBOM)

Bangalore, India

- Built and maintained **security-hardened curated container images** compliant with STIG/FIPS standards, eliminating vulnerable dependencies and minimizing runtime attack surface for cloud workloads.
- Designed and maintained **Helm charts** for Kubernetes deployments, enabling reproducible, configurable, and secure containerized application delivery.
- Automated container build, patching, testing, and release pipelines using Python and Bash, supporting RapidFort's zero-CVE curated image ecosystem.
- Implemented **secure software supply chain practices**, including **SBOM, reproducible builds, and CVE remediation** for production-ready container artifacts.

SIEMENS

MAY 2025 – AUG 2025

Software Developer Intern (Python, Flutter, NodeJS, MQTT, OPC-UA, FastMCP)

Bangalore, India

- Built a Model Context Protocol (**MCP**) client that connected with MCP servers controlling robotic arms, autonomous vehicles, and inventory systems. Used **FastMCP** and **FastAPI** to expose the client as an HTTP endpoint for smooth communication.
- Created a real-time **dashboard** using **Server-Sent Events (SSE)** to show live progress of each system task and allow users to control the MCP client directly from the interface.
- Developed a mobile app that takes PDF-based manual workflows and converts them into clear, step-by-step interactive guides, reducing operational labor requirements by nearly 50%.
- Worked with communication protocols like **OPC-UA** and **MQTT** to let different automation systems share data quickly and reliably.

SSquad Ventures

JAN 2025 – APR 2025

Full Stack Developer Intern (Android)

Remote/India

- Engineered a scalable social commerce platform **Outgoingg** with Flutter (frontend) and Node.js/Express (backend), enabling users to socialize, plan, shop, and manage advertisements through a unified interface.
- Optimized backend performance via **query indexing, caching, and pagination**, achieving a documented 35% reduction in API latency and ensuring robust data consistency across services.
- Implemented best practices in full-stack development including **state management, modular architecture, secure token handling, and CI/CD workflows** for maintainable and high-performing releases.

PROJECTS

Hypervisor Based Malware Defense System | Powershell, HyperDbg, Python, VM, Intel VT-x **GitHub**

- Developed a stealthy, hypervisor-based malware analysis engine operating at Ring -1 (VMX root) via HyperDbg to invisibly intercept 24 Windows kernel syscalls, utilizing live kernel structure traversal and EPROCESS side-channels for real-time metadata extraction.
- Built a fully automated, stateless detonation pipeline orchestrating 12-step per-sample workflows (including VM reversion, FakeNet simulation, and named-pipe synchronization), enabling unattended dynamic analysis and zero-install, in-VM PE arming without host filesystem exposure.

Multi-threaded Proxy server | C, Multithreading & Sockets, TCP/IP

GitHub

- Architected a multithreaded HTTP proxy server in C, efficiently managing 400+ simultaneous TCP connections using POSIX threads, semaphores, and non-blocking I/O.
- Implemented full HTTP/1.1 compliance with persistent connections, CONNECT tunneling, and intelligent header rewriting for seamless client-server communication.

SKILLS & COURSEWORK

Languages: C/C++, Python, Dart, JavaScript, TypeScript, Bash

Frameworks & Libraries: FastAPI, Flutter, React.js, Express.js, Tailwind CSS

DevOps & Tools: Docker, Kubernetes, Git/GitHub, Postman, VS Code, VirtualBox, VMware Workstation, Linux/Ubuntu, DRAKVUF, CAPEv2, HyperDbg

Cloud & Databases: Firebase, MongoDB, MySQL, SQLite

API & Communication: REST APIs, WebSockets, SSE (Server-Sent Events)

Courses: Algorithms; Database Management System; Computer Networks; Operating Systems; Deep Learning; **Big Data Analytics** (Hadoop, HDFS, MapReduce, Spark, Kafka, Zookeeper); Big Data Security; Reinforcement Learning; Financial Analytics

ACHIEVEMENTS

- **Specialist on Codeforces** with peak rating of 1433. **Profile**
- Ranked in the top 2% nationally in JEE Mains, India's most competitive engineering entrance exam with over 900,000 candidates.